

PIYUSH SUDHIR DESHMUKH

 Piyush Deshmukh |  piyush.deshmukh.in@gmail.com |  +91 9359918418 |  piyushd12

SKILLS

Frameworks	PyTorch, Keras, OpenCV, LangChain, LangGraph, FastMCP
Programming	Python, C/C++, SQL
Core Concepts	Data Structures & Algorithms, Object-Oriented Programming (OOP), Computer Networks

PROJECTS

- **Infrastructure Change Detection in GeoSpatial Imagery** Github
 - Developed a deep learning based change detection system to identify and analyze damaged buildings by comparing pre and post disaster satellite images.
 - Utilized **DINOv2** for feature extraction and **VM-UNet** for semantic segmentation, fine-tuned on the **xBD** dataset for accurate damage classification.
 - Generated multi-color-coded heatmaps to visually represent the severity and extent of damage across affected areas.
- **Timetable Generation using Genetic Algorithm** Github
 - Designed a genetic algorithm in **Python** to generate optimal class and lab schedules under strict time and resource constraints.
 - Implemented fitness functions to avoid conflicts and maximize timetable compactness and fairness.
 - Enabled easy customization for number of days, time slots, courses, and departments.
- **Sign Language Translation using Sequence Learning** Github
 - Developed a deep learning pipeline to translate sign gestures into text using **MediaPipe** hand tracking and **LSTM** networks.
 - Built a custom dataset collection tool for users to record and label their own sign gestures.
 - Achieved real-time inference using **TensorFlow**; supports generalization to any custom or regional sign language.

EDUCATION

SIES Graduate School of Technology, Navi Mumbai	2022 - present
Bachelor of Engineering, Information Technology	CGPA: 9.01 / 10
Sundarrao Solanke Mahavidyalaya	2020 - 2022
Higher Secondary Certificate (HSC)	Percentage: 80.67 / 100

PUBLICATIONS

- Paper titled “**Hybrid Foundation-State Space Networks for Building Damage Assessment in Remote Sensing**” selected for an **Oral Presentation** at CVR-2026.

ACHIEVEMENTS

- Secured **2nd Runner-up** position at the **GenAI Showdown Hackathon**.
- Selected as a **Regionalist** in **Code Uncode 2025**, a national-level competitive programming contest.

CERTIFICATIONS

- Earned the **OCI Generative AI Professional** certification from Oracle Cloud Infrastructure, covering foundations and practical applications of Generative AI.